

M351 - PROBABILITY I

Introduction

Probability theory is a branch of mathematics that focuses on analyzing and quantitatively assessing "chance" or "randomness".

Indeed, many phenomena in real-life produce outcomes that are observable (after occurring), but not predictable with certainty (before occurring).

Example: . tomorrow's weather

. outcome of tossing a die

. gambling (or "games of chance")

Hence the need to calculate "likelihoods", "risks", "chances" in many fields such as finance (stock market), engineering, physics, biology, chemistry, medicine, genetics, pharmacology, epidemiology, etc.

One has typically an "intuitive" understanding of probability.

Example:

"If an unbiased coin is flipped, then there is 50% chance (on likelihood) of Tails coming up."

However a subjective interpretation of probability can vary from person to person.

"therefore"

∴ Need to use an axiomatic approach to develop a consistent/rigorous theory (however intuition does play a role in setting axioms, solving/interpreting problems).

This course provides a basic introduction to probability theory.

Review of Basic Set Theory

Definition: A set A is a collection of objects.

Key Idea: idea of membership to a set.

Examples: . set $\mathbb{Z}$ of all integers

. set of all letters in the alphabet

. set $\mathbb{R}$ of all real numbers

. set of all atoms in the universe

The objects in a set A are called elements of A .

Notation: $a \in A$ means "a is an element of set A "
or "a is a member of set A "
or "a belongs to set A "

$a \notin A$ means "a is not an element of set A "

Often one defines a set by listing its elements within curly brackets:

$$A = \{a, b, c, \dots, z\} \quad \text{each element listed once}$$

$$B = \{2\} \quad \leftarrow \text{singleton set}$$

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

$$D = \{0, 1, 4, 9, 16, 25, 36, \dots\}$$

Note: $\{a, b\} = \{b, a\} = \{a, a, b, b, b\}$

↑
the order of elements
in a set does not matter

↑ listing elements repeatedly
does not matter

More generally, if $p(x)$ is a statement describing the property about elements x of a set A , then we can write A as:

$$A = \{x : p(x)\}$$

↑
such that

i.e., A is the set of all objects x for which $p(x)$ holds.

Example:

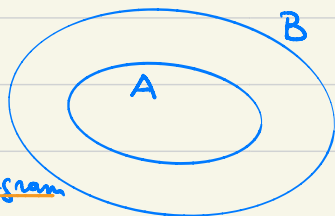
- $A = \{x \in \mathbb{Z} : x \geq 3\} = \{3, 4, 5, \dots\}$ is the set of all integers greater or equal to 3.
- $B = \{x \in \mathbb{R} : 0 \leq x \leq 1\} = [0, 1]$ ↙ the unit interval is the set of all real numbers between 0 and 1.
- $C = \{0, 1, 4, 9, 16, 25, 36, \dots\} = \{m : m = s^2, s \in \mathbb{Z}\}$ is the set of all square integers.

Definition Empty set

The empty set is a set with no elements and is denoted by $\emptyset$ or $\{\}$.

Subsets: Given sets A and B , we say that A is a subset of B and write $A \subset B$ (or $A \subseteq B$) if every element of A is also an element of B .

$A \subset B$ means if $x \in A$, then $x \in B$
or $\overset{\uparrow}{A} \subseteq B$



Venn diagram

Examples

$$\bullet \quad \mathbb{N} \subset \mathbb{Z} \subset \mathbb{R} \subset \mathbb{C}$$

$\{0, 1, 2, 3, \dots\}$ set of natural numbers
 $\uparrow$ set of integers
 $\uparrow$ set of real numbers
 $\uparrow$ set of complex numbers $= \{z: z = x + iy, x, y \in \mathbb{R}\}$
where $i^2 = -1$ (i imaginary unit)

$\bullet \quad \mathbb{Z} \not\subset \mathbb{N} \rightarrow$ not a subset

$\bullet$ For any set A , $\emptyset \subset A$

Set Equality: Two sets A and B are equal if they have exactly the same elements. We write: $A = B$

Principle of Set Equality:

$$A = B \iff A \subset B \text{ and } B \subset A$$

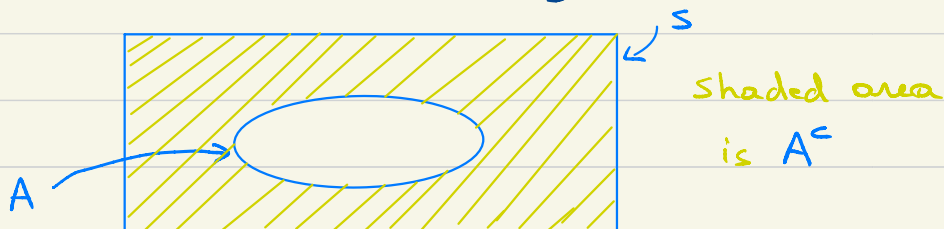
↑
"is equivalent to"
or "if and only if"

Universe:

We will consider throughout subsets of a set S , called the universe or the universal set.

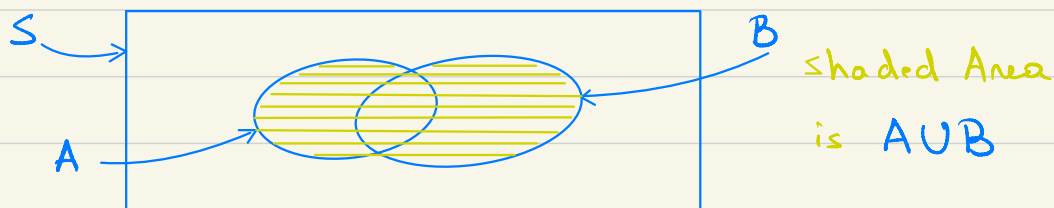
Complementation: Given universe S and $A \subset S$, then the complement of A (in S), denoted by A^c , is the set of all elements of S that are not in A :

$$A^c = \{x \in S : x \notin A\}$$



Set Union: Given two subsets A and B in universe S, the union of A and B, denoted by $A \cup B$, is the set of all elements which are in A, in B or both:

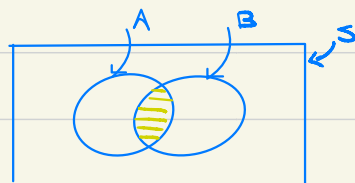
$$A \cup B = \{x \in S : x \in A \text{ or } x \in B \text{ (or both)}\}$$



Set Intersection: Given subsets A and B in S, the intersection of A and B, denoted by $A \cap B$ or AB , is the set of all elements that belong to both sets:

$$AB = \{x \in S : x \in A \text{ and } x \in B\}$$

$\nearrow$
 $A \cap B$

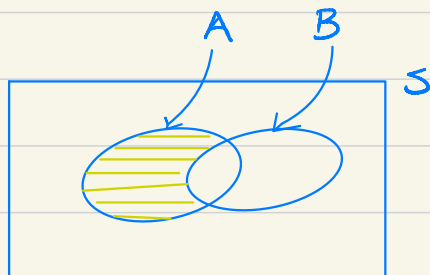


Definition: Sets A and B are said to be (mutually) disjoint or mutually exclusive if they have no elements in common:

$$AB = \emptyset$$

Set Difference: Given subsets A and B in S, the difference of A and B, denoted by $A - B$ (or $A \setminus B$) is the set of elements of A that are not in B:

$$A - B = \{x \in A : x \notin B\} = AB^c$$



shaded area is $A - B$

Note:

- $S - A = A^c$
- In general, $A - B \neq B - A$

Cartesian Products:

The Cartesian product of sets A and B is the set $A \times B$ defined as follows:

$$A \times B = \{ (a, b) : a \in A \text{ and } b \in B \}$$

Example:

$$A = \{1\} \text{ and } B = \{1, 2\}$$

then

$$A \times B = \{ (1, 1), (1, 2) \}$$

but

$$B \times A = \{ (1, 1), (2, 1) \}$$

Notation: $A \times A = A^2$, $\underbrace{A \times A \times \dots \times A}_{n \text{ times } (n \geq 1)} = A^n$

Properties of Set Operations

- $S^c = \emptyset$ and $\emptyset^c = S$
- $(A^c)^c = A$
- $A \cup \emptyset = A$ and $A \cap \emptyset = \emptyset$
- $A \cup S = S$ and $A \cap S = A$
- $A \cup A^c = S$ and $A \cap A^c = \emptyset$
- Commutativity

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

- Associativity

$$A \cup (B \cap C) = (A \cup B) \cap C$$

$$A \cap (B \cup C) = (A \cap B) \cup C$$

- Distributivity

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

- De Morgan's Laws

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$